



Data Visualization Associate Early Remote Internship

**Team 16 – Week 1 Report
Dataset Overview, Data Quality & EDA**

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**Submission Date:
Monday, 12th January 2026**

Week 1 deliverables

Applicant Dataset

Dataset Overview & Documentation of Applicant Data

Dataset Size

- **Total rows:** Approximately **6,895 applicant records** (each row represents one applicant, including the header).
- **Total columns:** **30 variables**

Types of Variables

The dataset contains a mix of categorical, numerical, date/time, identifier, and free-text variables.

Categorical Variables

- **Given_Name** – Categorical (Text)
Applicant's first name.
- **Last_Name** – Categorical (Text)
Applicant's last name.
- **Major** – Categorical
Intended or applied academic program.
- **Country** – Categorical
Country of residence or origin.
- **University** – Categorical
University to which the applicant has applied.
- **Citizenship** – Categorical
Applicant's citizenship.
- **Intake** – Categorical
Academic intake term (e.g., Fall 2024, Summer 2024).
- **College_1st_Choice** – Categorical
Applicant's first-choice college or school.
- **City** – Categorical
City of residence.
- **Region** – Categorical
State, province, or regional location.
- **Application_Source** – Categorical
Application submission platform (e.g., Slate, INTO, ERX Web App).
- **Most_Recent_Released_Decision** – Categorical
Latest admission decision status.
- **Comments** – Categorical (Free text)
Additional remarks or notes related to the application.

- **Is_Global_Grad** – Categorical (Yes/No)
Indicates whether the applicant is classified as a global graduate.

Numerical Variables (Identifiers)

- **Phone_Number** – Numeric (Identifier)
Primary contact phone number.
- **Phone_Number_2** – Numeric (Identifier)
Secondary contact phone number.

Categorical Variables Stored in Numeric Format

- **Postal** – Categorical
Postal or ZIP code stored as numeric values.

Date / Time Variables

- **Date_of_Birth** – Date
Applicant's date of birth.
- **Admit_Date** – Date
Date on which an admission decision was recorded.
- **Created_At** – DateTime
Timestamp when the applicant record was created.
- **Modified_At** – DateTime
Timestamp of the most recent modification to the record.
- **Recieved_At** – DateTime
Timestamp indicating when the application was received.

Identifier Variables

- **Reference_ID** – Numeric (Identifier)
Unique internal identifier for each applicant.
- **SEVIS_ID** – Text (Identifier)
Immigration identifier for international students.
- **Application_Agency_Code** – Text (Identifier)
Code representing the recruitment agency or partner.

Contact and Address Fields

- **Street_1** – Text
Primary street address.
- **Street_2** – Text
Secondary address line.
- **Street_3** – Text
Additional address details.
- **Email_ID** – Text
Personal email address.

- **Official_University_email_address** – Text
Institution-issued email address.

General Dataset Themes

- The dataset represents international and domestic student application records.
- It captures demographic, academic, geographic, and administrative information.
- The data supports admissions decision-making, intake planning, and international student tracking.
- Multiple application pipelines are represented, reflecting diverse recruitment channels.

Initial Observations & Notable Characteristics

- A significant proportion of applicants originate from India, Ghana, Nigeria, and other international regions.
- Several fields contain NULL or missing values, especially in address and contact information.
- Phone numbers and timestamps are often stored in scientific notation, requiring data cleaning before analysis.
- Intake terms span multiple academic periods, including Summer, Fall, and Spring.
- The dataset contains substantial personally identifiable information (PII) and must be handled with care.

Data Usage Considerations

- Suitable for admissions analytics, demographic analysis, and recruitment reporting.
- PII fields should be masked or anonymized for analytical and presentation purposes.
- Identifier fields such as Reference_ID should be used for joining with Connect, Outreach, and SEVIS datasets.
- Date fields should be validated and standardized before time-based analysis.
- Categorical variables may require label normalization across systems prior to modeling or visualization.

Connect Dataset

Dataset Overview & Documentation of Connect Data

Dataset Size

- **Total rows:** Approximately **34,341 records** (each row represents one applicant record in the Connect system, including the header).
- **Total columns:** **8 variables**

Types of Variables

The dataset contains a mix of categorical, date/time, and identifier variables.

Categorical Variables

- **Deposit_Status** – Categorical
Indicates whether an applicant has paid, pending, or not paid the enrollment deposit.
- **Assigned** – Categorical
Represents assignment status (e.g., counselor assigned, unassigned, or processing stage).
- **University** – Categorical
Identifies the university or institution associated with the applicant.
- **I_20_Status** – Categorical
Indicates the status of the I-20 document (e.g., Issued, Pending, Not Started).

Date / Time Variables

- **Received_At** – DateTime
Timestamp indicating when the applicant record was received in the Connect system.
- **Created_At** – DateTime
Timestamp showing when the record was created in the system.
- **Modified_At** – DateTime
Timestamp capturing the most recent update to the record.

Identifier Variables

- **Reference_ID** – Numeric (Identifier)
Unique identifier used to link applicant records across systems such as admissions and Connect platforms.

General Dataset Themes

- The dataset represents post-application processing and enrollment tracking data.
- It focuses on application progression, deposit confirmation, and immigration documentation status.
- The dataset supports international admissions workflows, particularly I-20 issuance and enrollment readiness.
- Records appear to be synchronized with broader applicant systems using a shared Reference_ID.

Initial Observations & Notable Characteristics

- The dataset contains a large volume of records, indicating active tracking across multiple admission cycles.
- Deposit_Status and I_20_Status are key operational variables for monitoring applicant readiness.
- Timestamp fields enable process timeline analysis, such as time between receipt and I-20 issuance.
- Some records may contain missing or delayed status updates, especially in assignment or deposit fields.

- The dataset includes minimal personal identifiable information (PII) compared to applicant datasets.
- When merged with other datasets, appropriate privacy and access controls should be applied.

Data Usage Considerations

- Suitable for enrollment funnel analysis, operational monitoring, and international student reporting.
- Date/time fields may require format standardization before analysis.
- Reference_ID should be validated for relationship integrity when joining datasets.
- Status fields may need category standardization prior to visualization or modeling.

Outreach Dataset

Dataset Overview & Documentation of Outreach Data

Dataset Size

- **Total rows:** Approximately **multiple thousands of outreach records** (each row represents one outreach interaction or activity related to an applicant, including the header).
- **Total columns:** **Multiple variables** capturing outreach details, timing, and status.

Types of Variables

The dataset contains a mix of **categorical**, **date/time**, **identifier**, and **free-text** variables.

Categorical Variables

- **Outreach_Type** – Categorical
Indicates the type of outreach activity (e.g., email, call, follow-up, campaign).
- **Outreach_Channel** – Categorical
Represents the medium used for outreach (e.g., Email, Phone, System Notification).
- **Outreach_Status** – Categorical
Indicates the status of the outreach (e.g., Completed, Pending, Failed).
- **Assigned** – Categorical
Shows whether the outreach activity is assigned to a counselor or staff member.
- **University** – Categorical
Identifies the institution associated with the outreach activity.
- **Comments / Notes** – Categorical (Free text)
Contains additional remarks or notes entered by staff regarding the outreach.

Date / Time Variables

- **Outreach_Date** – DateTime
Timestamp indicating when the outreach activity occurred or was scheduled.
- **Created_At** – DateTime
Records when the outreach entry was created in the system.

- **Modified_At** – DateTime
Captures the most recent update made to the outreach record.
- **Received_At** – DateTime
Indicates when the outreach record was received or logged in the system.

Identifier Variables

- **Reference_ID** – Numeric (Identifier)
Unique identifier used to associate outreach activities with individual applicants across datasets.
- **Outreach_ID** – Numeric or Text (Identifier)
Unique identifier for each outreach interaction.

General Dataset Themes

- The dataset represents communication and engagement activities conducted with prospective or admitted students.
- It supports tracking of follow-ups, campaigns, and applicant engagement efforts.
- The data is designed to complement applicant and connect datasets by providing interaction-level insights.
- Outreach records can occur multiple times per applicant, creating a one-to-many relationship with applicant data.

Initial Observations & Notable Characteristics

- Multiple outreach records may exist for a single applicant, reflecting ongoing engagement.
- Outreach channels and types allow analysis of communication effectiveness.
- Free-text comments may contain inconsistent or unstructured information, requiring text cleaning for analysis.
- Some outreach records may have missing statuses or assignment values, particularly for automated campaigns.
- Timestamp fields enable analysis of response timing and engagement patterns.
- The dataset may indirectly relate to PII when joined with applicant data and should be handled carefully.

Data Usage Considerations

- Suitable for engagement analysis, communication performance tracking, and process monitoring.
- One-to-many joins should be carefully managed when linking to applicant-level datasets using Reference_ID.
- Date/time variables may require timezone and format standardization.
- Categorical fields should be reviewed for label consistency before reporting or visualization.
- Free-text fields may require text preprocessing if used for qualitative analysis.

SEVIS Dataset

Dataset Overview & Documentation of SEVIS Data

Dataset Size

- **Total rows:** Approximately **several thousand SEVIS records** (each row represents one SEVIS-related record linked to an applicant, including the header).
- **Total columns: Multiple variables** capturing SEVIS identifiers, status, and compliance-related details.

Types of Variables

The dataset contains a mix of categorical, date/time, identifier, and status variables.

Categorical Variables

- **SEVIS_Status** – Categorical
Indicates the current SEVIS status of the applicant (e.g., Active, Initial, Terminated, Completed).
- **Visa_Type** – Categorical
Represents the visa category associated with the SEVIS record (e.g., F-1, J-1).
- **University** – Categorical
Identifies the institution responsible for the SEVIS record.
- **Program_Level** – Categorical
Indicates the academic level of the student (e.g., Undergraduate, Graduate).
- **Remarks / Comments** – Categorical (Free text)
Contains additional notes related to SEVIS processing or compliance actions.

Date / Time Variables

- **SEVIS_Created_Date** – Date
Date on which the SEVIS record was initially created.
- **SEVIS_Updated_Date** – Date
Date reflecting the most recent update to the SEVIS record.
- **Program_Start_Date** – Date
Indicates the official start date of the academic program.
- **Program_End_Date** – Date
Indicates the expected or actual end date of the academic program.
- **Created_At** – DateTime
Timestamp showing when the SEVIS record was created in the system.
- **Modified_At** – DateTime
Timestamp capturing the most recent modification to the SEVIS record.
- **Recieved_At** – DateTime
Timestamp indicating when the SEVIS information was received or logged.

Identifier Variables

- **SEVIS_ID** – Text (Identifier)
Unique government-issued identifier assigned to international students for immigration tracking.
- **Reference_ID** – Numeric (Identifier)
Internal identifier used to link SEVIS records to applicant, connect, and outreach datasets.
- **Record_ID** – Numeric or Text (Identifier)
Unique identifier for each SEVIS record within the dataset.

General Dataset Themes

- The dataset represents immigration and regulatory compliance information for international students.
- It supports tracking of visa status, program timelines, and SEVIS lifecycle events.
- The data is essential for federal reporting, compliance monitoring, and institutional oversight.
- SEVIS records are typically linked to applicant data through shared identifiers.

Initial Observations & Notable Characteristics

- Each applicant is typically associated with one primary SEVIS record, though updates may generate multiple entries over time.
- SEVIS_Status is a critical variable for monitoring legal enrollment eligibility.
- Date fields enable analysis of program duration and compliance timelines.
- Some records may have missing or delayed updates, depending on reporting schedules.
- Free-text remarks may vary in structure and content, requiring cleaning for analysis.
- The dataset contains highly sensitive regulatory identifiers and must be handled securely.

Data Usage Considerations

- Suitable for compliance reporting, immigration monitoring, and audit support.
 - SEVIS_ID must be protected and anonymized in analytical or reporting outputs.
 - Joins with other datasets should use Reference_ID rather than SEVIS_ID when possible.
 - Date variables should be validated for logical consistency (e.g., start dates before end dates).
 - Access to this dataset should be restricted to authorized users only due to regulatory sensitivity.
-

Report of SEVIS Dataset

Total missing values: 196455

Columns with missing values: 70/123

Columns with missing values:

	Missing_Count	Missing_Percentage
Current_Session_End_Date	3852	100.00
Thesis_Or_Dissertation	3852	100.00
Termination_Reason	3852	100.00
SEVIS_Legacy_Last_Name	3852	100.00
SEVIS_Legacy_Middle_Name	3852	100.00
SEVIS_Legacy_First_Name	3852	100.00
Other_Termination_Comment	3852	100.00
Suffix	3852	100.00
Minor_Description	3852	100.00
Cancellation_Reason	3852	100.00
Study_Research_Abroad	3852	100.00
Last_Session	3852	100.00
Next_Session_Start_Date	3852	100.00
US_Mailing_Address_2	3852	100.00
US_Mailing_Routing_Code	3851	99.97
US_Physical_Address_2	3847	99.87
US_Telephone	3847	99.87
Other_Costs_Comment	3846	99.84
Passport_Name	3843	99.77
US_Mailing_Address_Status	3840	99.69
US_Mailing_Address_1	3840	99.69
US_Mailing_City	3840	99.69
US_Mailing_State	3840	99.69
US_Mailing_Postal_Code	3840	99.69
US_Physical_Routing_Code	3839	99.66
Date_of_Departure	3835	99.56
Port_of_Departure	3835	99.56
English_Is_Not_Required_Because	3831	99.45
Foreign_Telephone_Country_Code	3828	99.38
Foreign_Telephone	3828	99.38
US_Physical_Address_1	3822	99.22
US_Physical_Address_Status	3822	99.22
US_Physical_State	3822	99.22
US_Physical_City	3822	99.22
US_Physical_Postal_Code	3822	99.22
Major_2_Description	3820	99.17
City_of_Birth	3762	97.66
I_94_Admission_Number	3747	97.27
Passport_Country_of_Issuance	3732	96.88
Passport_Expiration_Date	3732	96.88
Port_of_Entry	3730	96.83
Date_of_Entry	3730	96.83
FIN_ID	3730	96.83
Foreign_State_Province	3639	94.47
Visa_Issuance_Post	3475	90.21
Visa_Expiration_Date	3328	86.40
Visa_Issuance_Date	3327	86.37
Visa_Number	3321	86.21
Passport_Number	3320	86.19
Foreign_Address_2	2082	54.05
I-901_Fee_Payment_Receipt	1744	45.28
I-901_Transaction_Date	1744	45.28
I-901_Transaction_Type	1744	45.28
I-901_Transaction_Amount	1744	45.28
Foreign_Postal_Code	704	18.28
Other_Source_Type	685	17.78
School_Fund_Type	528	13.71
Overall_Remarks	402	10.44
Given_Name	150	3.89

On-Campus_Employment	16	0.42
Other_Costs	13	0.34
Dependent_Expenses	12	0.31
Funds_From_Other_Sources	6	0.16
Foreign_City	6	0.16
Email	4	0.10
Funds_From_This_School	3	0.08
Border_Commuter_Student	3	0.08
Student_Has_English_Proficiency	2	0.05
No_Phone_Indicator	1	0.03
Date_of_Birth	1	0.03

Now we will drop the columns having null percentage up to 70%.

Remaining columns now

	Missing_Count	Missing_Percentage
Foreign_Address_2	2082	54.05
I-901_Fee_Payment_Receipt	1744	45.28
I-901_Transaction_Amount	1744	45.28
I-901_Transaction_Date	1744	45.28
I-901_Transaction_Type	1744	45.28
Foreign_Postal_Code	704	18.28
Other_Source_Type	685	17.78
School_Fund_Type	528	13.71
Overall_Remarks	402	10.44
Given_Name	150	3.89
On-Campus_Employment	16	0.42
Other_Costs	13	0.34
Dependent_Expenses	12	0.31
Foreign_City	6	0.16
Funds_From_Other_Sources	6	0.16
Email	4	0.10
Funds_From_This_School	3	0.08
Border_Commuter_Student	3	0.08
Student_Has_English_Proficiency	2	0.05
No_Phone_Indicator	1	0.03
Date_of_Birth	1	0.03

Duplicate record checks

Full row duplicates: 0 (0.0%)

'SEVIS_ID': 0 duplicates, 3852 unique values

'NonImmigrant_ID': 0 duplicates, 3852 unique values

Outliers or unusual entries

Found outliers in 3 numeric columns:

- Funds_From_Other_Sources: 1209 outliers (31.4%)
Range: 0.00 to 250000.00, Mean: 34115.17
- Funds_From_This_School: 501 outliers (13.0%)
Range: 0.00 to 76964.00, Mean: 5717.86
- NonImmigrant_ID: 88 outliers (2.3%)
Range: 17555070.00 to 36047456.00, Mean: 35561803.35

Dropped 25 columns as they contained the same value

DROP: 'Student_Requested_Reinstatement' (constant value, no analytical value)
 DROP: 'School_Code' (constant value, no analytical value)
 DROP: 'CPT_Approved' (constant value, no analytical value)
 DROP: 'Student_Transfer_Out' (constant value, no analytical value)
 DROP: 'Student_Authorized_to_Drop_Below_Full_Course_of_Study' (constant value, no analytical value)
 DROP: 'STEM_OPT_Requested' (constant value, no analytical value)
 DROP: 'Class_of_Admission' (constant value, no analytical value)
 DROP: 'Length_of_Next_Break_of_Vacation' (constant value, no analytical value)
 DROP: 'Border_Commuter_Student' (constant value, no analytical value)
 DROP: 'Major_1_CIP_2020_Conversion' (constant value, no analytical value)
 DROP: 'Campus_Name' (constant value, no analytical value)
 DROP: 'Campus_Code' (constant value, no analytical value)
 DROP: 'OCE_Approved' (constant value, no analytical value)
 DROP: 'Student_Extension_Requested' (constant value, no analytical value)
 DROP: 'Pre-Completion_OPT_Approved' (constant value, no analytical value)
 DROP: 'Student_Change_of_Status_Request' (constant value, no analytical value)
 DROP: 'OCE_Requested' (constant value, no analytical value)
 DROP: 'Pre-Completion_OPT_Requested' (constant value, no analytical value)
 DROP: 'Correction_Request_Pending' (constant value, no analytical value)
 DROP: 'Post-Completion_OPT_Approved' (constant value, no analytical value)
 DROP: 'School_Name' (constant value, no analytical value)
 DROP: 'STEM_OPT_Approved' (constant value, no analytical value)
 DROP: 'Post-Completion_OPT_Requested' (constant value, no analytical value)
 DROP: 'SEVIS_Status' (constant value, no analytical value)
 DROP: 'Cap_Gap_Extended' (constant value, no analytical value)

Now final dataset contains shape: (3852, 49)

Exploratory Data Analysis (EDA)

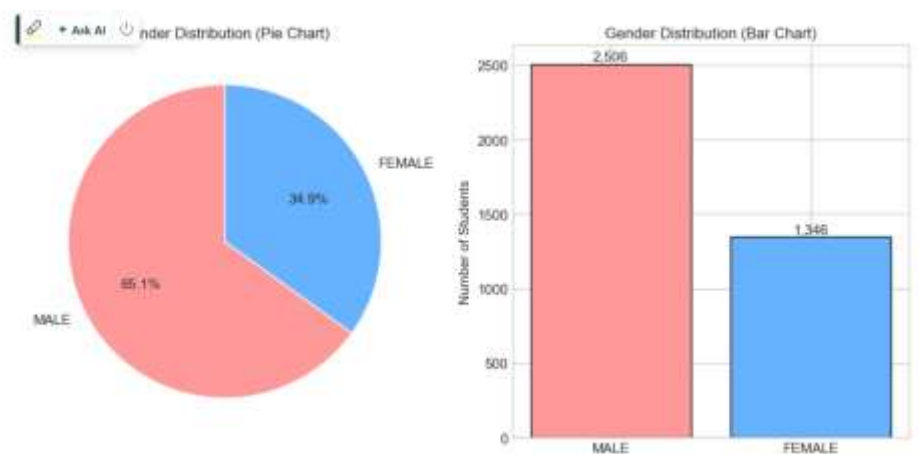
DATASET OVERVIEW

- Total Records: 3,852 students
- Features: 49 columns
- Data Types:
 - object: 35 columns
 - float64: 8 columns
 - int64: 6 columns

DEMOGRAPHIC ANALYSIS

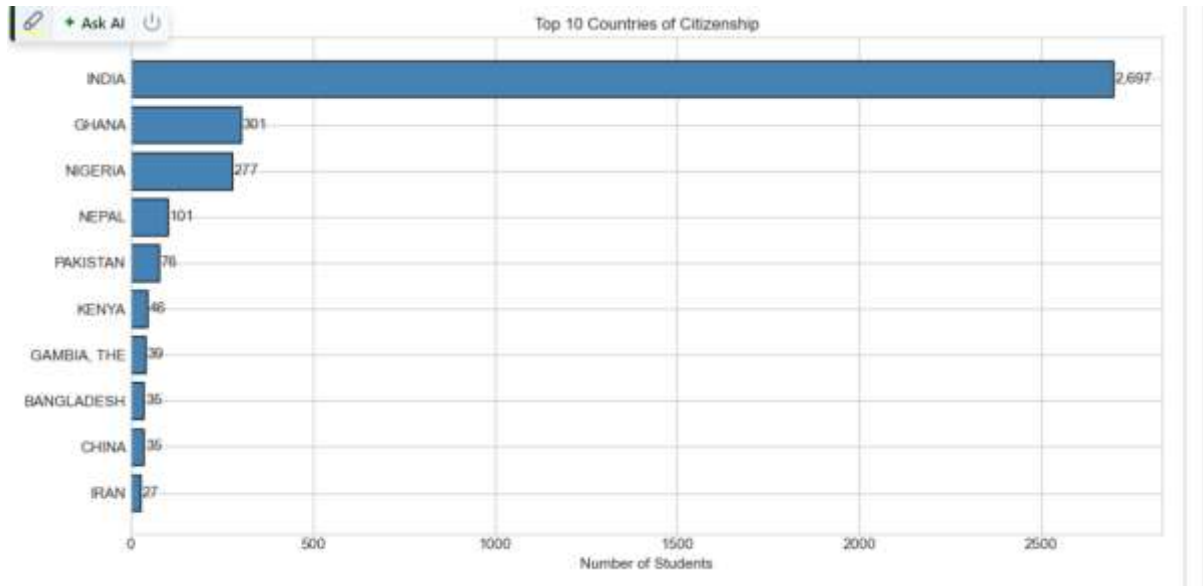
Gender Distribution:

- MALE: 2,506 (65.1%)
- FEMALE: 1,346 (34.9%)



Country of Citizenship (Top 10):

- INDIA: 2,697 (70.0%)
- GHANA: 301 (7.8%)
- NIGERIA: 277 (7.2%)
- NEPAL: 101 (2.6%)
- PAKISTAN: 76 (2.0%)
- KENYA: 46 (1.2%)
- GAMBIA, THE: 39 (1.0%)
- BANGLADESH: 35 (0.9%)
- CHINA: 35 (0.9%)
- IRAN: 27 (0.7%)

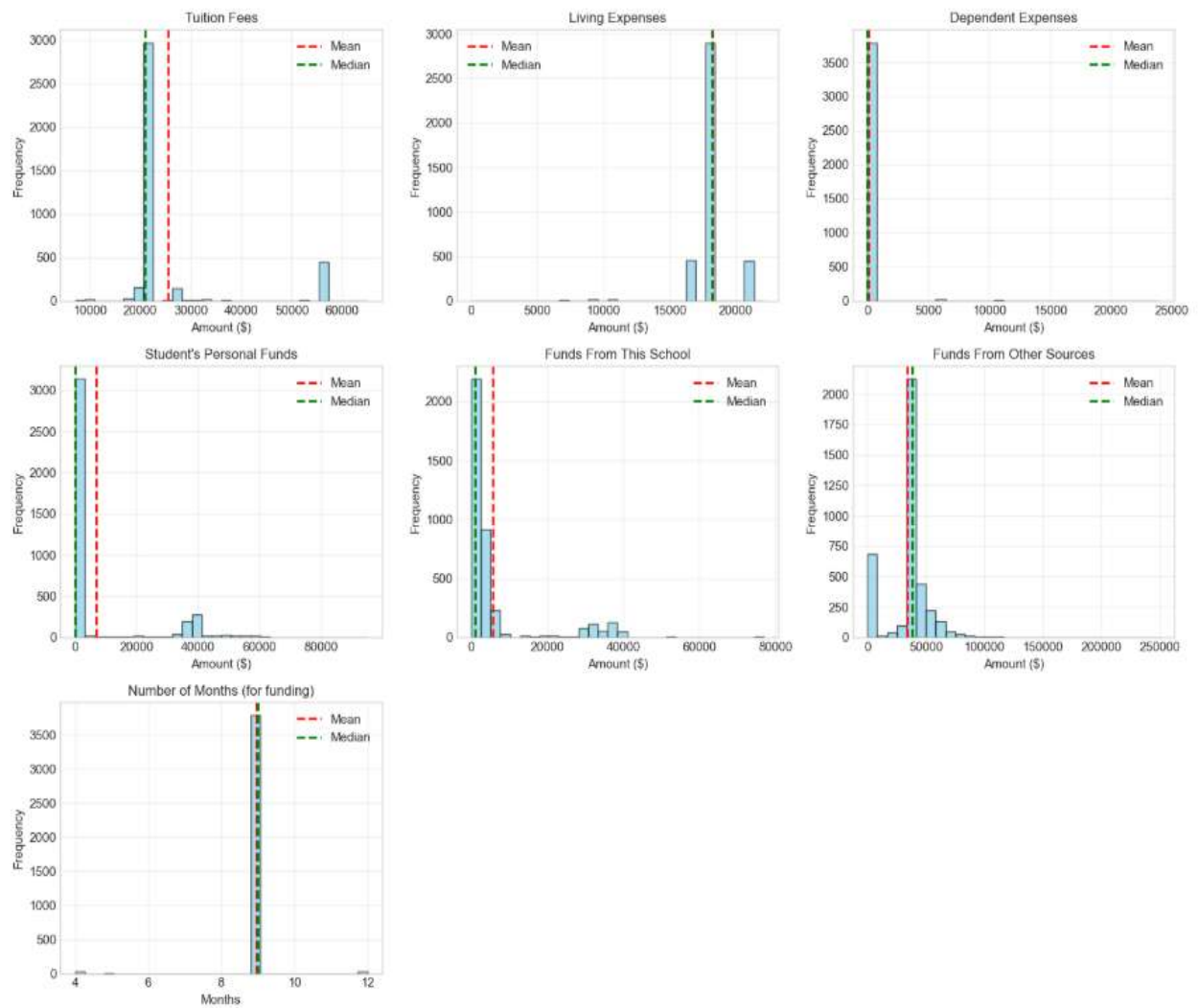


. FINANCIAL ANALYSIS

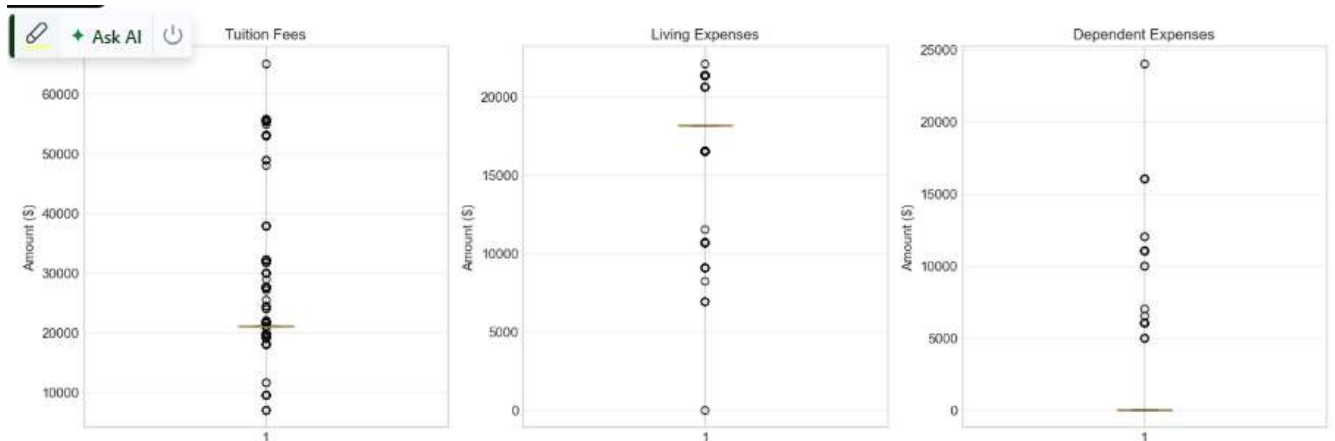
Financial Summary Statistics

Metric	Count	Mean	Median	Std	Min	Max
Tuition Fees	3852	\$25,514	\$21,000	\$11,405	\$7,055	\$65,000
Living Expenses	3852	\$18,231	\$18,150	\$1,546	\$0	\$22,100
Dependent Expenses	3840	\$110	\$0	\$1,087	\$0	\$24,000
Student's Personal Funds	3852	\$6,970	\$0	\$15,365	\$0	\$95,000
Funds From This School	3849	\$5,718	\$1,000	\$11,117	\$0	\$76,964
Funds From Other Sources	3846	\$34,115	\$38,150	\$18,626	\$0	\$250,000
Number of Months (for funding)	3852	9.0	9.0	0.6	4.0	12.0

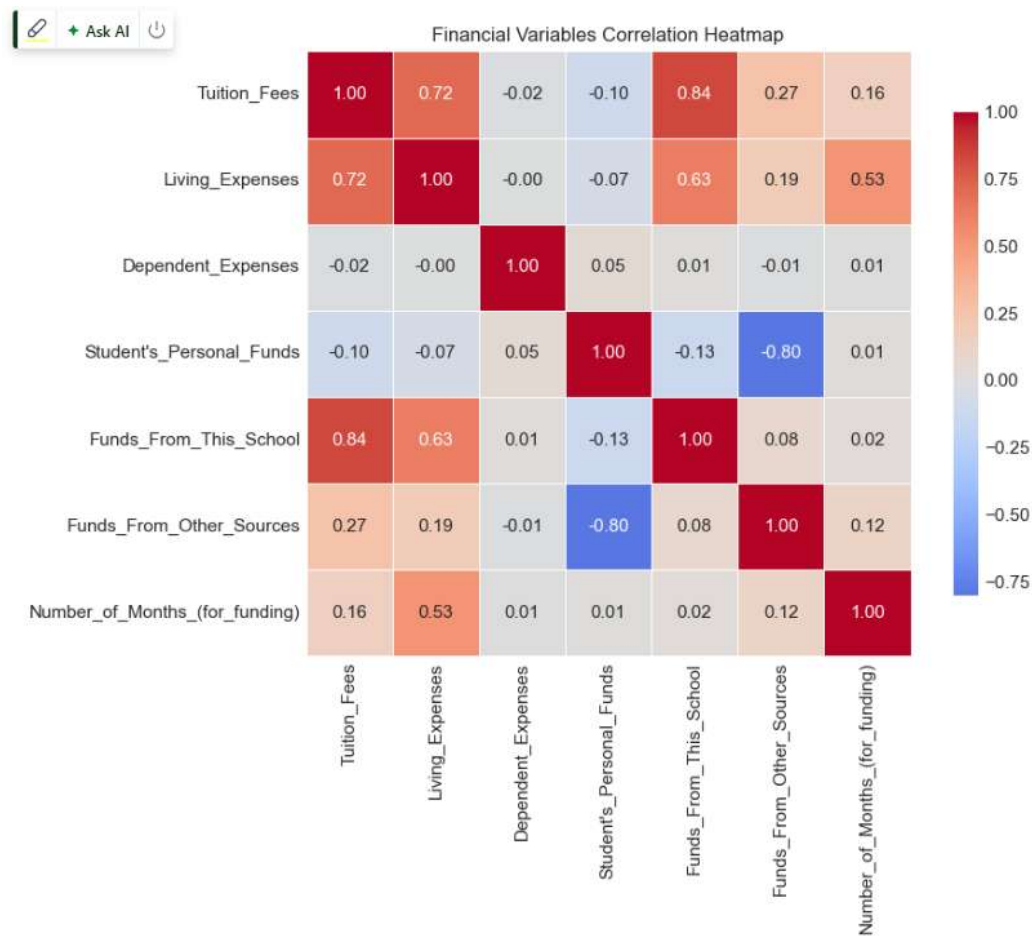
Financial Distributions:



Outlier Detection (Boxplots):



Correlation Analysis:



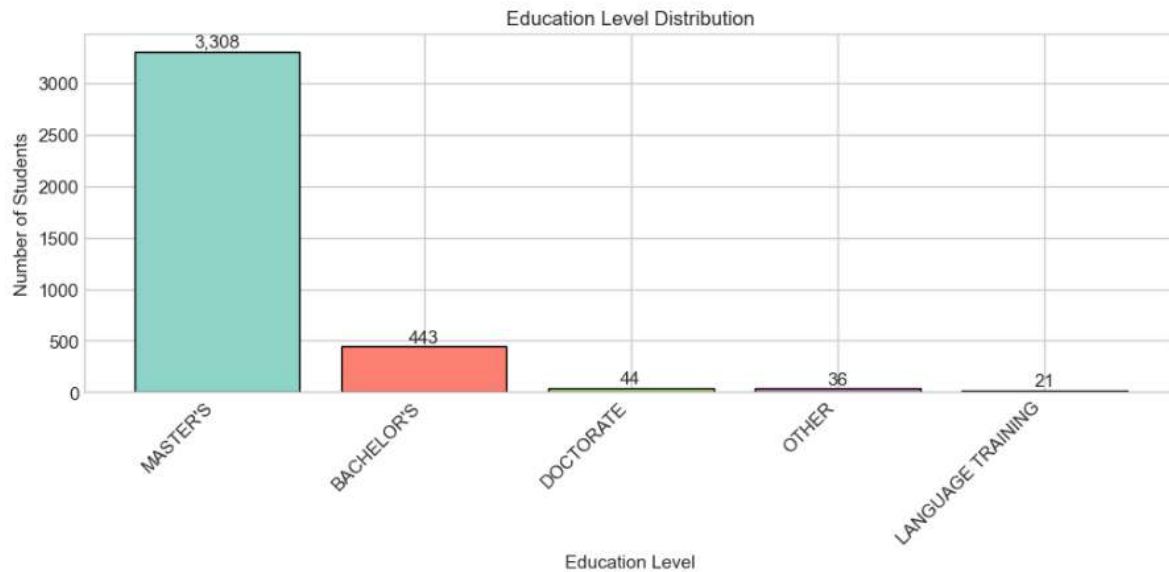
Strong Correlations (> 0.5):

- Tuition_Fees ↔ Living_Expenses: 0.718
- Tuition_Fees ↔ Funds_From_This_School: 0.838
- Living_Expenses ↔ Funds_From_This_School: 0.631
- Living_Expenses ↔ Number_of_Months_(for_funding): 0.527
- Student's_Personal_Funds ↔ Funds_From_Other_Sources: -0.801

ACADEMIC ANALYSIS

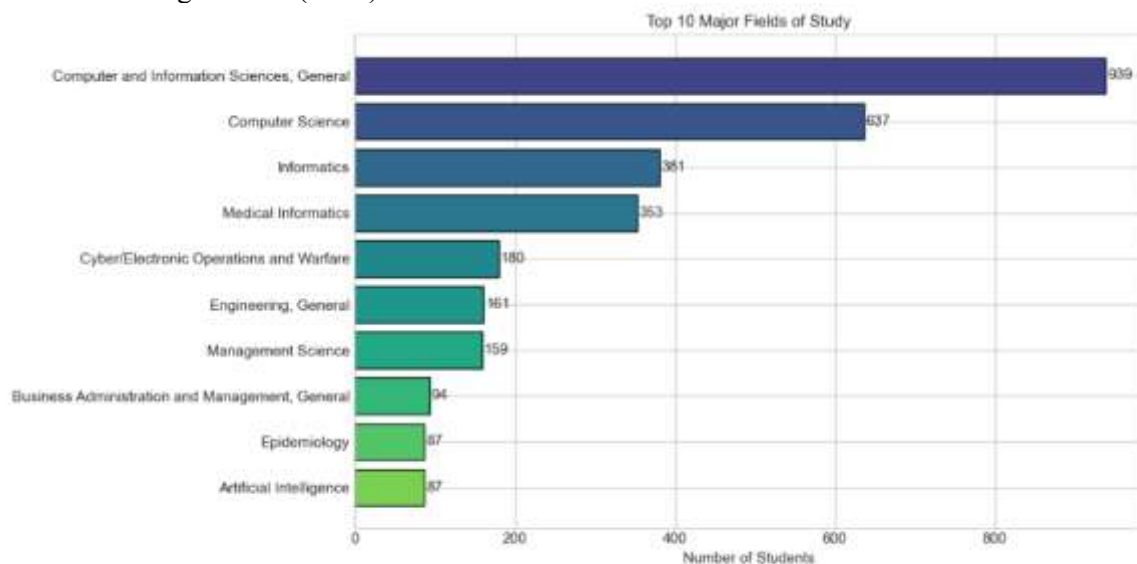
Education Level Distribution:

- MASTER'S: 3,308 (85.9%)
- BACHELOR'S: 443 (11.5%)
- DOCTORATE: 44 (1.1%)
- OTHER: 36 (0.9%)
- LANGUAGE TRAINING: 21 (0.5%)



Top 10 Majors:

- Computer and Information Sciences, General: 939 (24.4%)
- Computer Science: 637 (16.5%)
- Informatics: 381 (9.9%)
- Medical Informatics: 353 (9.2%)
- Cyber/Electronic Operations and Warfare: 180 (4.7%)
- Engineering, General: 161 (4.2%)
- Management Science: 159 (4.1%)
- Business Administration and Management, General: 94 (2.4%)
- Epidemiology: 87 (2.3%)
- Artificial Intelligence: 87 (2.3%)



MISSING VALUE ANALYSIS

Total missing values: 11,591

Columns with missing values: 20/49

Top 10 Columns with Missing Values:

	Missing_Count	Missing_Percentage
Foreign_Address_2	2082	54.05
I-901_Fee_Payment_Receipt	1744	45.28
I-901_Transaction_Amount	1744	45.28

I-901_Transaction_Date	1744	45.28
I-901_Transaction_Type	1744	45.28
Foreign_Postal_Code	704	18.28
Other_Source_Type	685	17.78
School_Fund_Type	528	13.71
Overall_Remarks	402	10.44
Given_Name	150	3.89

MAJOR FINDINGS:

1. DEMOGRAPHIC PATTERNS:

- Gender distribution shows MALE students are the majority
- INDIA is the most represented country (2,697 students)
- Top country represents 70.0% of all students

2. FINANCIAL PATTERNS:

- Tuition fees show \$11,405 standard deviation
- Average tuition: \$25,514
- 81.3% of students report \$0 personal funds

3. ACADEMIC PATTERNS:

- MASTER'S is the most common education level
- Students are distributed across 87 different majors

4. DATA QUALITY:

- 20 columns contain missing values
- 29 columns are complete
- Overall data completeness: 93.9%

Report of Outreach Dataset

Dataset Shape: 1565 rows × 18 columns

MISSING VALUE ANALYSIS

Top 15 columns with missing values:

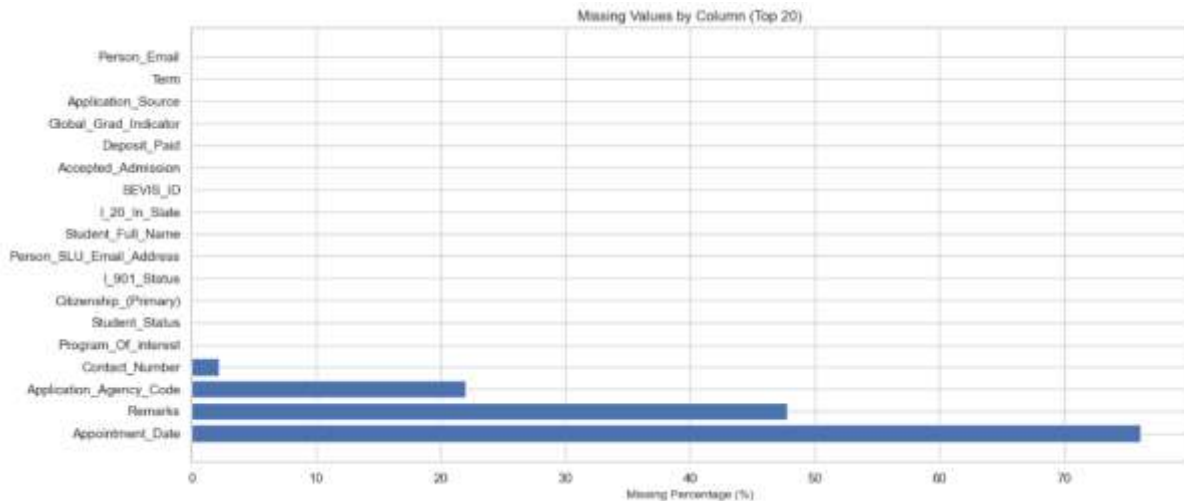
column_name	missing_count	missing_percentage	data_type
Appointment_Date	1192	76.166134	object
Remarks	748	47.795527	object
Application_Agency_Code	344	21.980831	object
Contact_Number	34	2.172524	object
Program_Of_interest	0	0.000000	object
Student_Status	0	0.000000	object
Citizenship_(Primary)	0	0.000000	object
I_901_Status	0	0.000000	object
Person_SLU_Email_Address	0	0.000000	object
Student_Full_Name	0	0.000000	object
I_20_In_Slate	0	0.000000	object
SEVIS_ID	0	0.000000	object
Accepted_Admission	0	0.000000	object
Deposit_Paid	0	0.000000	object
Global_Grad_Indicator	0	0.000000	object

Missing Value Summary:

Total missing cells: 2,318

Average missing percentage: 8.23%

Columns with >50% missing: 1



So dropping Appointment_Date

DUPLICATE RECORD ANALYSIS

Complete duplicate rows: 0 (0.00%)

DATA TYPE CONSISTENCY CHECK

Columns with potential mixed data types: ['SEVIS_ID', 'Contact_Number']

BASIC STATISTICS

Numeric columns (0): []

Categorical columns (18): ['Student_Full_Name', 'Program_Of_interest', 'Term', 'Application_Source', 'Global_Grad_Indicator', 'Deposit_Paid', 'Accepted_Admission', 'SEVIS_ID', 'I_20_In_Slate', 'Person_Email']...

CATEGORICAL VARIABLES OVERVIEW

Unique values in categorical columns

	unique_count
Term	2
Global_Grad_Indicator	2
Deposit_Paid	2
Accepted_Admission	2
I_20_In_Slate	2
Application_Source	7
Program_Of_interest	47
Student_Full_Name	1558
Person_Email	1560
SEVIS_ID	1561

High cardinality columns (>50 unique values): ['Student_Full_Name', 'Person_Email', 'SEVIS_ID']

OUTLIERS IN TEXT LENGTH FIELDS

Name_Length:

- Total values: 1565
- Normal range: [4.0, 36.0]
- Outliers detected: 84 (5.4%)
- Outlier values: [37, 40, 41, 43, 45]

Email_Length:

- Total values: 1565
- Normal range: [15.5, 35.5]
- Outliers detected: 30 (1.9%)
- Outlier values: [36, 37, 38, 41, 42]

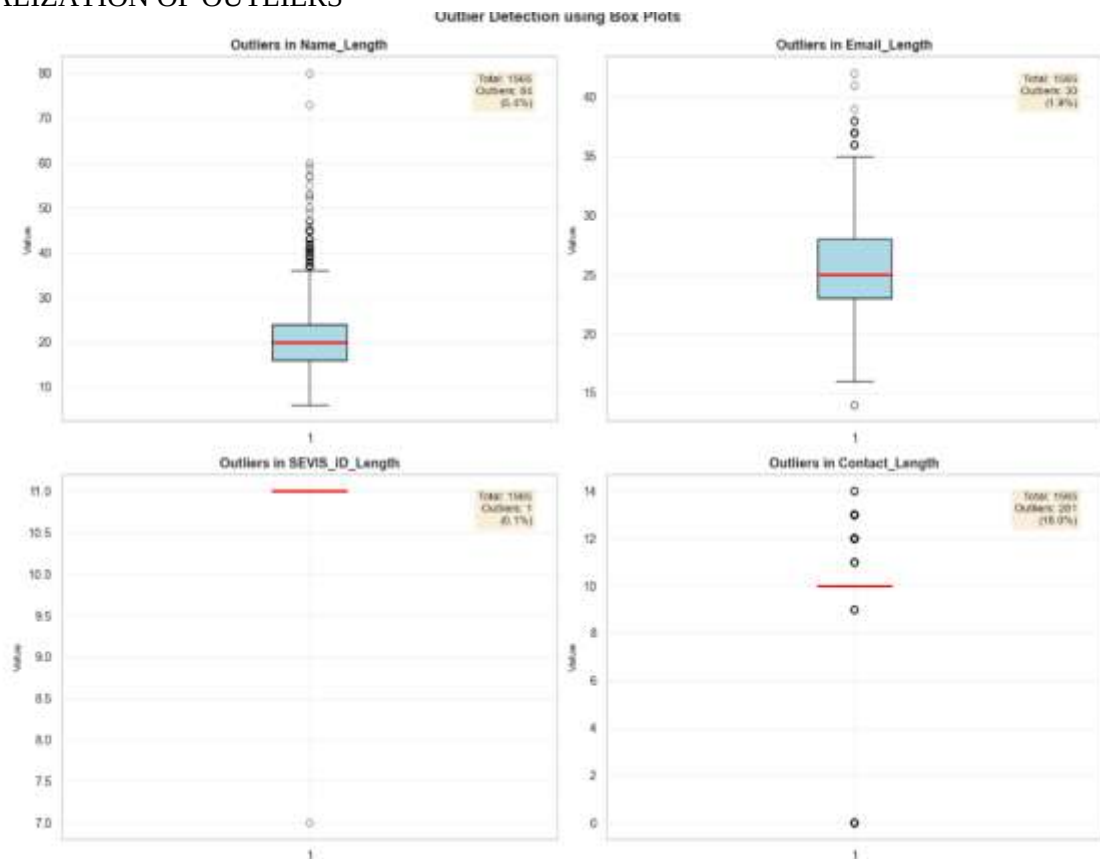
SEVIS_ID_Length:

- Total values: 1565
- Normal range: [11.0, 11.0]
- Outliers detected: 1 (0.1%)
- Outlier values: [7]

Contact_Length:

- Total values: 1565
- Normal range: [10.0, 10.0]
- Outliers detected: 281 (18.0%)
- Outlier values: [0, 9, 12, 13, 14]

VISUALIZATION OF OUTLIERS



UNUSUAL CATEGORIES IN CATEGORICAL DATA

Program_Of_interest:

- Total categories: 47
- Rare categories (<1% frequency): 32
- Top rare categories:
 - 'Computer Science (PhD)': 8 occurrences (0.51%)
 - 'Software Engineering (MS)': 8 occurrences (0.51%)
 - 'Communication (MA)': 7 occurrences (0.45%)

Citizenship_(Primary):

- Total categories: 34
- Rare categories (<1% frequency): 29
- Top rare categories:
 - 'Nepal': 15 occurrences (0.96%)
 - 'Bangladesh': 13 occurrences (0.83%)
 - 'Kenya': 13 occurrences (0.83%)

Application_Source:

- Total categories: 7
- Rare categories (<1% frequency): 4
- Top rare categories:
 - 'SOPHAS': 11 occurrences (0.70%)
 - 'CollegeNet': 8 occurrences (0.51%)
 - 'Element': 2 occurrences (0.13%)

CONTACT NUMBER VALIDITY ANALYSIS

Contact Number Length Distribution:

- 0 digits: 37 numbers (2.4%) - UNUSUAL
- 9 digits: 10 numbers (0.6%) - UNUSUAL
- 10 digits: 1284 numbers (82.0%) - VALID
- 11 digits: 14 numbers (0.9%) - UNUSUAL
- 12 digits: 125 numbers (8.0%) - UNUSUAL
- 13 digits: 90 numbers (5.8%) - UNUSUAL
- 14 digits: 5 numbers (0.3%) - UNUSUAL

Unusual Contact Number Lengths Detected:

- 0 digits: 37 numbers
Sample: [nan, nan, nan]
- 9 digits: 10 numbers
Sample: ['901018844', '547891175', '508778515']
- 13 digits: 90 numbers
Sample: ['2347067025690', '2347030221910', '2348039552191']
- 14 digits: 5 numbers
Sample: ['88001861478364', '23408069590727', '44447552246661']

EMAIL ADDRESS QUALITY ANALYSIS

Email Address Validity Check:

- Valid email format: 1564 (99.9%)
- Potentially invalid: 1 (0.1%)

Unusual Email Domains (rare domains):

- aol.com: 2 occurrences
- alemayohu.samuelbrhane.com: 1 occurrences

- protonmail.com: 1 occurrences
- stu.ucc.edu.gh: 1 occurrences
- my.stlcc.edu: 1 occurrences

OUTLIER IMPACT SUMMARY

Outlier Counts by Field (IQR method):

- Contact_Length: 281 outliers (18.0%)
- Name_Length: 84 outliers (5.4%)
- Email_Length: 30 outliers (1.9%)
- SEVIS_ID_Length: 1 outliers (0.1%)

EDA REPORT

A.DATASET OVERVIEW

Total Records: 1,565

Total Columns: 30

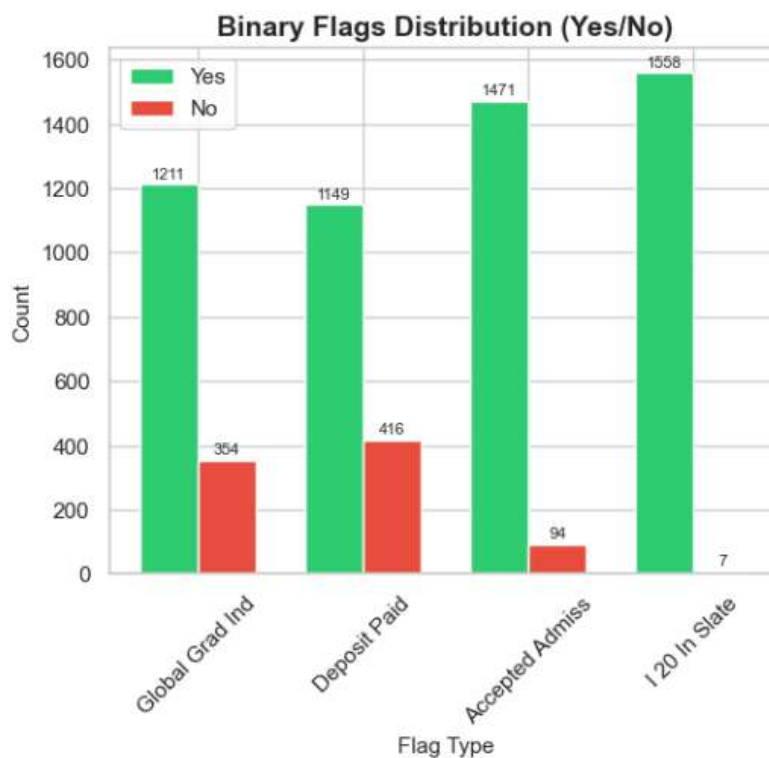
Total Students: 1,558

Total Unique Email IDs: 1,560

Total Unique SEVIS IDs: 1,561

BINARY FLAGS SUMMARY (Yes/No Columns)

	Yes	Yes%	No
Global_Grad_Indicator	1211	77.4%	354
Deposit_Paid	1149	73.4%	416
Accepted_Admission	1471	94.0%	94
I_20_In_Slate	1558	99.6%	7



CATEGORICAL COLUMNS - VALUE COUNTS

Term:

Fall 2024: 1563 (99.9%)

Fall 2 2024: 2 (0.1%)

Application_Source:

INTO: 782 (50.0%)

Slate: 629 (40.2%)

ERX Web APP: 132 (8.4%)

SOPHAS: 11 (0.7%)

CollegeNet: 8 (0.5%)

Element: 2 (0.1%)

NursingCAS: 1 (0.1%)

Citizenship_(Primary):

India: 1295 (82.7%)

Nigeria: 76 (4.9%)

Ghana: 52 (3.3%)

Pakistan: 25 (1.6%)

China: 16 (1.0%)

Nepal: 15 (1.0%)

Bangladesh: 13 (0.8%)

Kenya: 13 (0.8%)

Iran: 12 (0.8%)

Uganda: 6 (0.4%)

... and 24 more categories

I_901_Status:

Payment: 1565 (100.0%)

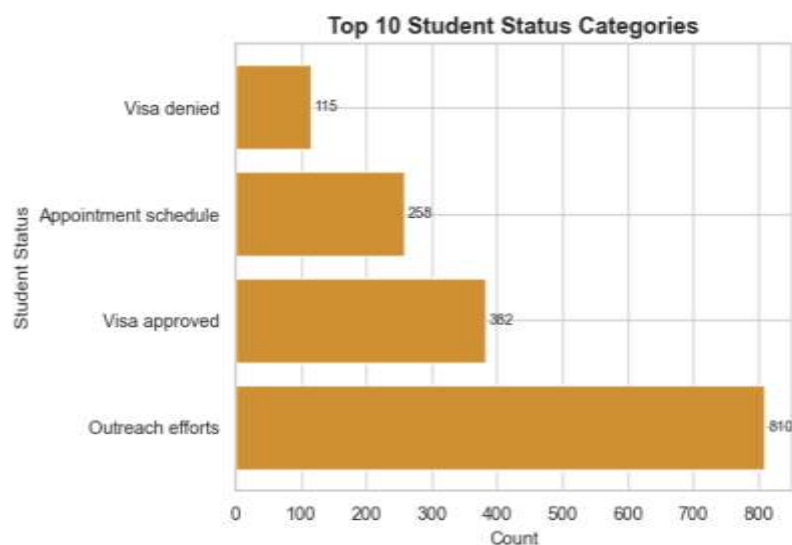
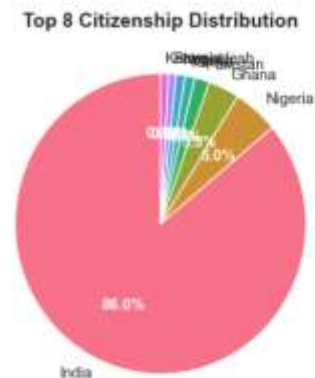
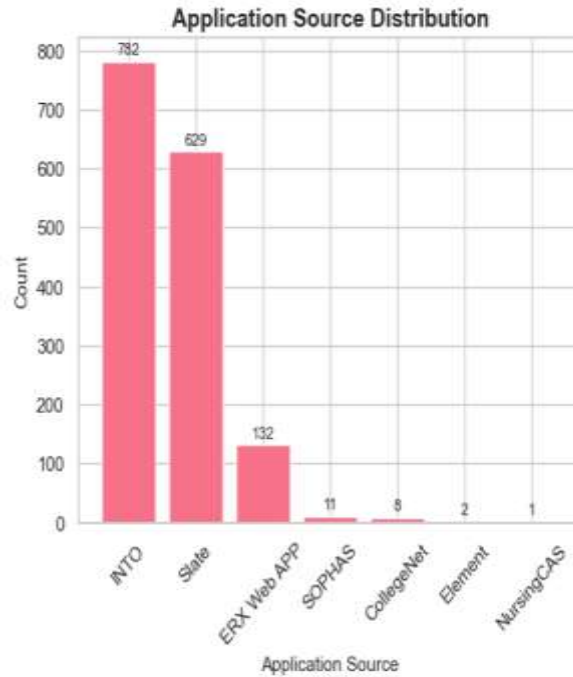
Student_Status:

Outreach efforts: 810 (51.8%)

Visa approved: 382 (24.4%)

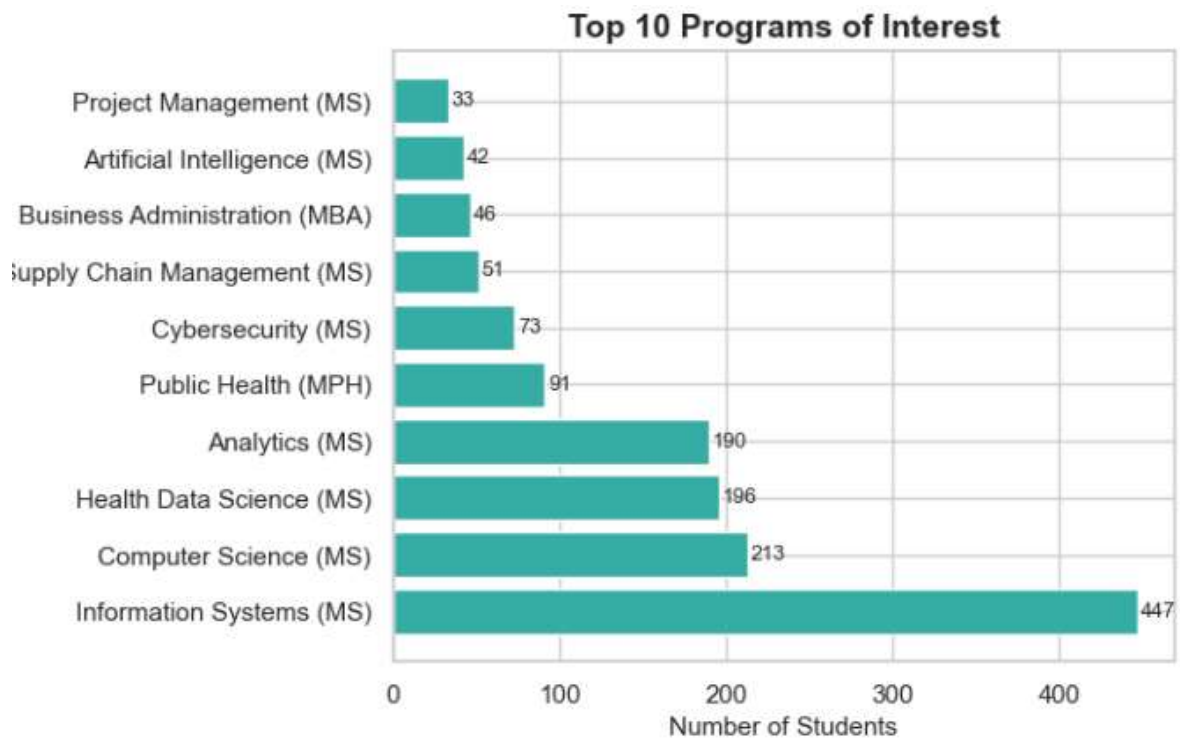
Appointment schedule: 258 (16.5%)

Visa denied: 115 (7.3%)



Program_Of_interest:

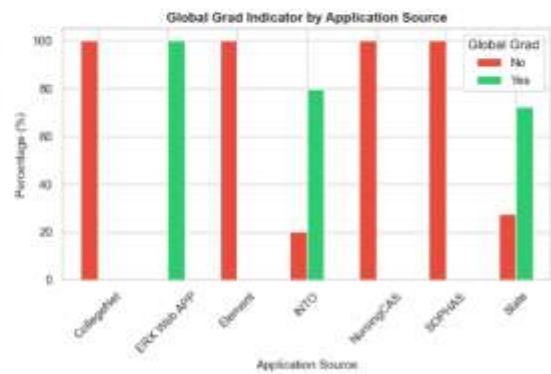
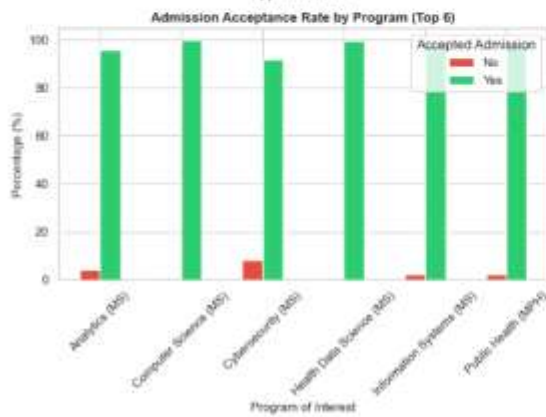
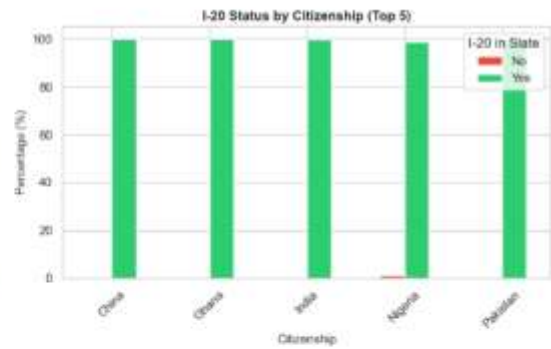
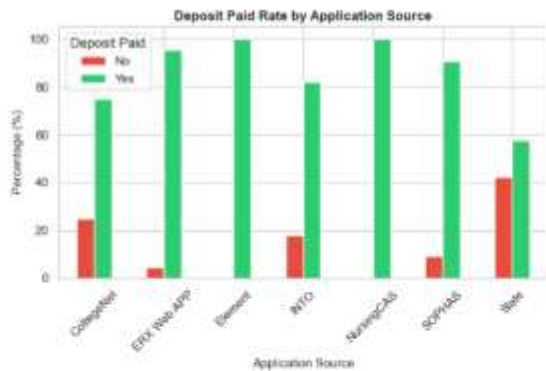
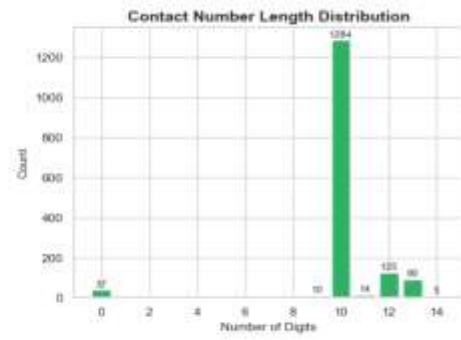
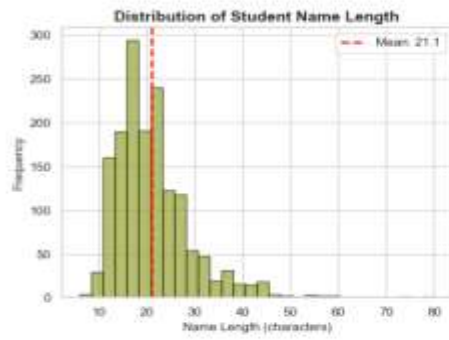
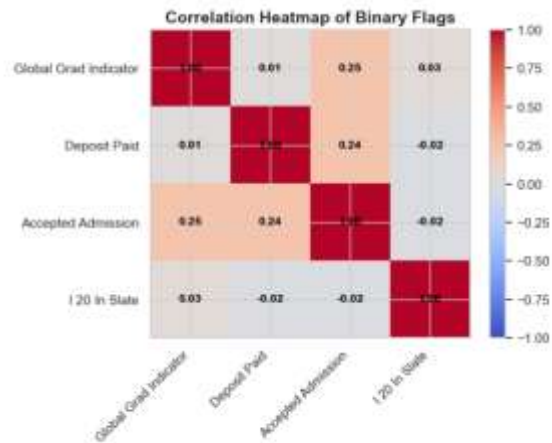
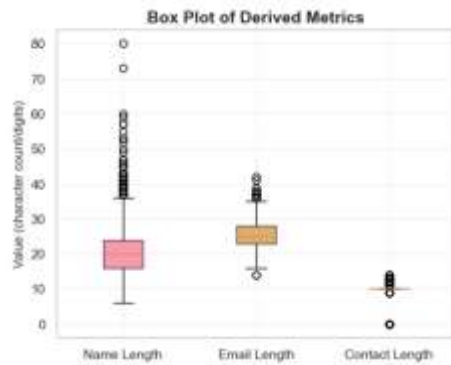
Information Systems (MS): 447 (28.6%)
Computer Science (MS): 213 (13.6%)
Health Data Science (MS): 196 (12.5%)
Analytics (MS): 190 (12.1%)
Public Health (MPH): 91 (5.8%)
Cybersecurity (MS): 73 (4.7%)
Supply Chain Management (MS): 51 (3.3%)
Business Administration (MBA): 46 (2.9%)
Artificial Intelligence (MS): 42 (2.7%)
Project Management (MS): 33 (2.1%)
... and 37 more categories



DERIVED METRICS SUMMARY

	Name_Length	Email_Length	SEVIS_ID_Length	Contact_Length
count	1565.00	1565.00	1565.0	1565.00
mean	21.12	25.68	11.0	10.11
std	7.91	4.12	0.1	1.80
min	6.00	14.00	7.0	0.00
25%	16.00	23.00	11.0	10.00
50%	20.00	25.00	11.0	10.00
75%	24.00	28.00	11.0	10.00
max	80.00	42.00	11.0	14.00

VISUALIZATIONS



KEY PATTERNS AND INSIGHTS

DATA PATTERNS OBSERVED:

1. Application Source Pattern: 'INTO' is the dominant source with 50.0% of applications
2. Citizenship Concentration: Top 3 countries account for 1423 students:
 - India: 1295 students (82.7%)
 - Nigeria: 76 students (4.9%)
 - Ghana: 52 students (3.3%)
3. Success Rate Metrics:
 - Deposit Paid: 73.4% Yes rate
 - Accepted Admission: 94.0% Yes rate
 - I 20 In Slate: 99.6% Yes rate
4. Program Popularity: Top 5 programs represent 1137 students:
 - Information Systems (MS): 447 students (28.6%)
 - Computer Science (MS): 213 students (13.6%)
 - Health Data Science (MS): 196 students (12.5%)
 - Analytics (MS): 190 students (12.1%)
 - Public Health (MPH): 91 students (5.8%)

DISTRIBUTION CHARACTERISTICS:

1. Name Length: Average 21.1 characters
 - Min: 6, Max: 80
 - Most common length: 15 characters
2. Contact Numbers: 1284 students have 10-digit numbers
 - Valid numbers: 1531 out of 1565
3. Student Status Diversity: 4 unique status categories
 - Most common: Outreach efforts (810 students)

CORRELATION INSIGHTS:

Correlations between key binary flags:

- Global Grad Indicator vs Deposit Paid: 0.010
- Global Grad Indicator vs Accepted Admission: 0.249
- Global Grad Indicator vs I 20 In Slate: 0.032
- Deposit Paid vs Accepted Admission: 0.244
- Deposit Paid vs I 20 In Slate: -0.019
- Accepted Admission vs I 20 In Slate: -0.017

Strongest correlation: Global_Grad_Indicator vs Accepted_Admission: 0.249

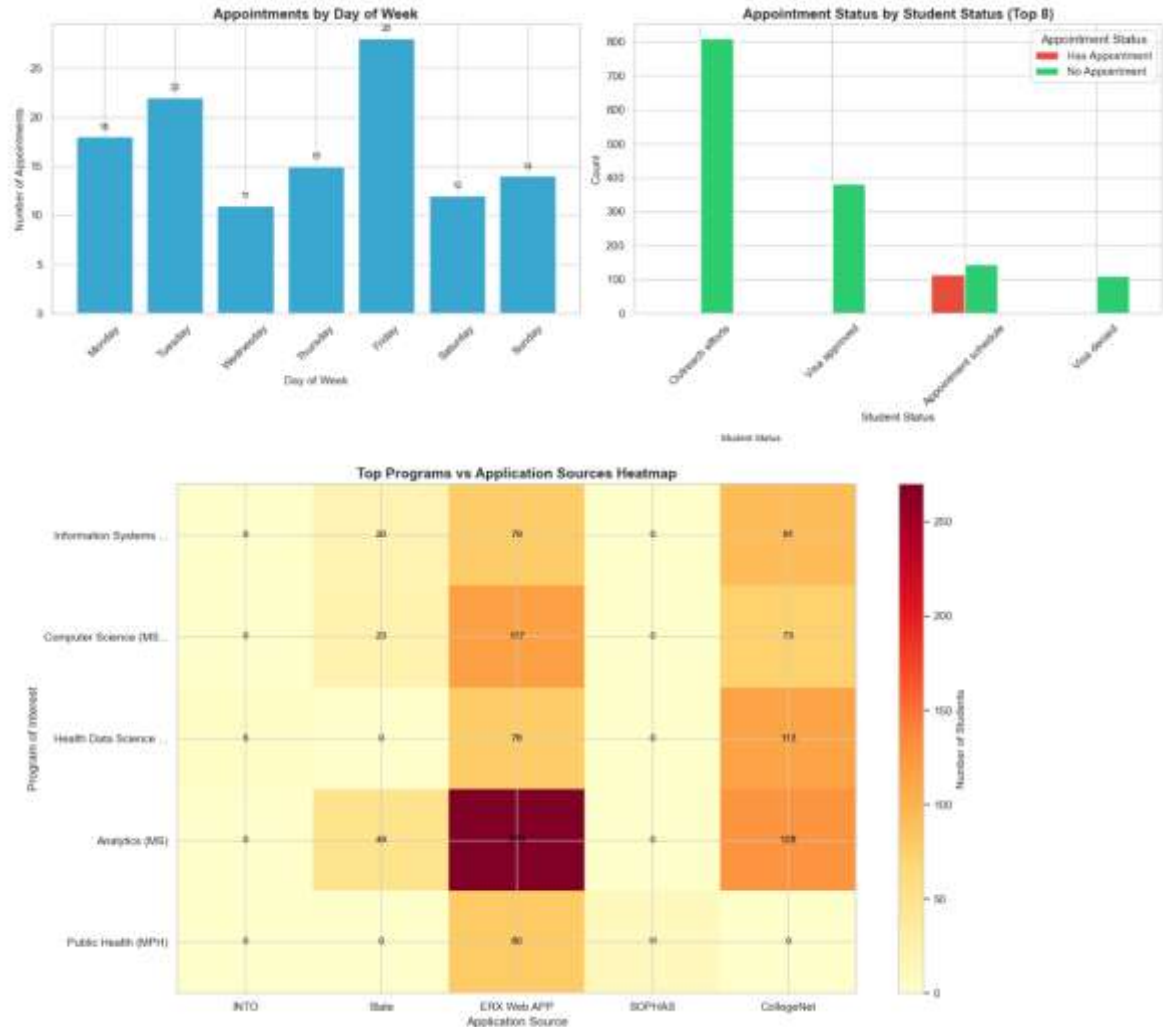
DATA QUALITY INSIGHTS:

1. Missing Data: Appointment_Date has 1192 missing values (76.2%)
2. Data Completeness: 28 out of 30 columns have <25% missing data
3. Unique Identifiers: All students have unique emails and SEVIS IDs
4. Data Consistency: Mixed data types found in SEVIS_ID and Contact_Number columns

INSIGHTS:

1. FOCUS AREAS:

- Concentrate outreach on top 5 programs which cover majority of students
- INSO SLU is the dominant application source - strengthen this channel
- India represents the largest student demographic - tailor communications



Report of APPLICANT Dataset

Dataset Shape: (6894, 30)

Total Records: 6894

Total Columns: 30

MISSING VALUE ANALYSIS

Missing values by column:

	Missing_Count	Missing_Percent	Data_Type
Admit_Date	6894	100.00	float64
Comments	6894	100.00	float64
Country	6875	99.72	object
College_1st_Choice	6875	99.72	object
Street_3	6831	99.09	object
Phone_Number_2	6704	97.24	object
Street_2	5165	74.92	object
Application_Agency_Code	4327	62.76	object
SEVIS_ID	4288	62.20	object
Postal	498	7.22	object

Region	319	4.63	object
Most_Recent_Released_Decision	146	2.12	object
Phone_Number	94	1.36	object
Official_University_email_address	63	0.91	object
Street_1	41	0.59	object
Is_Global_Grad	19	0.28	object
Date_of_Birth	19	0.28	object
City	5	0.07	object
Citizenship	2	0.03	object
Application_Source	1	0.01	object
Major	1	0.01	object

Total columns with missing values: 21

Total missing values: 56061

Overall missing percentage: 27.11%

Drop these columns as they have majority null values : ['Admit_Date','Comments','Country','College_1st_Choice','Street_3','Phone_Number_2','Street_2']

DUPLICATE RECORD ANALYSIS

Complete duplicate rows: 0

Checking duplicates in ID columns: ['Reference_ID', 'Email_ID', 'SEVIS_ID']

- Reference_ID: 1 duplicates

Duplicate values: [1324269]...

- Email_ID: 1 duplicates

Duplicate values: ['rebeccaagyare67@gmail.com']...

- SEVIS_ID: 4288 duplicates

Duplicate values: [nan 'N0036610616']...

Potential duplicate checks on: ['Given_Name','Last_Name','Phone_Number']

Duplicates based on first 2 key columns: 22

Original dataset shape: (6894, 23)

HANDLING DUPLICATES

Complete duplicates removed: 0

Reference_ID duplicates removed: 1

Email_ID duplicates removed: 0

Name-based duplicates removed: 22

Records after duplicate removal: 6871

HANDLING MISSING VALUES

Current missing values (after dropping high-missing columns):

	Missing_Count	Missing_Percent
Application_Agency_Code	4313	62.77
SEVIS_ID	4273	62.19
Postal	496	7.22
Region	318	4.63
Most_Recent_Released_Decision	146	2.12
Phone_Number	93	1.35
Official_University_email_address	63	0.92
Street_1	41	0.60
Is_Global_Grad	19	0.28
Date_of_Birth	19	0.28
City	5	0.07
Citizenship	2	0.03
Major	1	0.01
Application_Source	1	0.01

Missing value handling strategy:

Region: Filled 318 missing values with mode: 'Telangana'

Most_Recent_Released_Decision: Filled 146 missing values with mode: '8/28/2024'

Phone_Number: Filled 93 missing values with mode: '2.35E+12'

Official_University_email_address: Filled 63 missing values with mode: 'aachalreddy.nagireddy@slu.edu'

Street_1: Filled 41 missing values with mode: 'Accra'

Is_Global_Grad: Filled 19 missing values with mode: 'Yes'

Date_of_Birth: Filled 19 missing values with mode: '1/1/2001'

City: Filled 5 missing values with mode: 'Hyderabad'

Citizenship: Filled 2 missing values with mode: 'India'

Major: Filled 1 missing values with mode: 'Information Systems (MS)'

Application_Source: Filled 1 missing values with mode: 'Slate'

Postal: Filled 496 missing values with 'Unknown'

SEVIS_ID: Created missing flag and filled 4273 missing values with 'Not_Available'

Application_Agency_Code: Filled 4313 missing values with 'Not_Specified'

Total missing values after cleaning: 0

Percentage of missing values after cleaning: 0.00%

HANDLING OUTLIERS

Reference_ID outliers detected: 306

Lower bound: 1343408.25

Upper bound: 1520278.25

Outliers range: 407891.00 to 100004958.00

Outliers capped to bounds (new column 'Reference_ID_capped' created)

New columns added:

1. SEVIS_ID_missing - Flag column (1 if SEVIS_ID was missing, 0 otherwise)
2. Reference_ID_capped - Reference_ID with outliers capped
3. _datetime **columns** - For Recieved_At, Created_At, Modified_At (timestamp conversion)

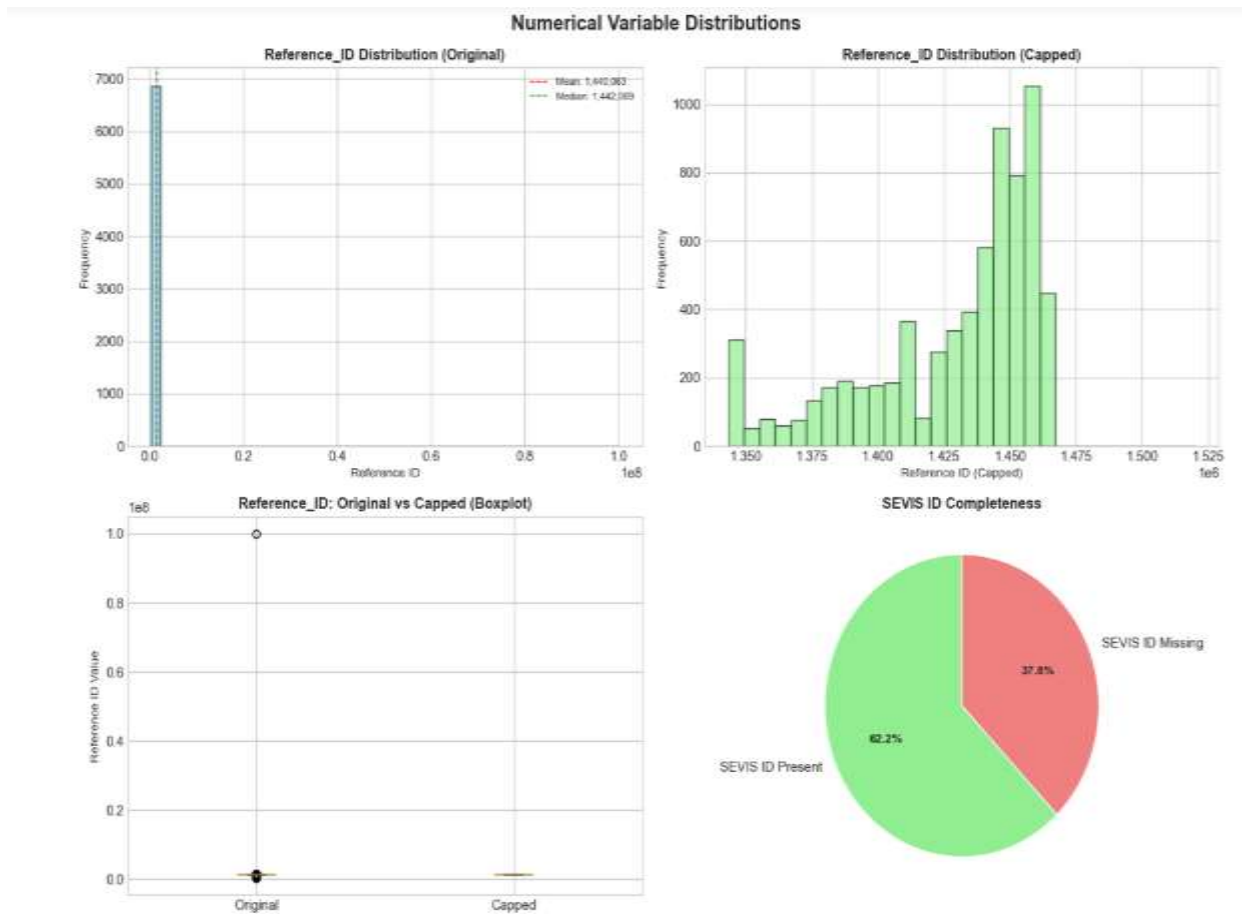
Final dataset shape after removing identical columns: (6871, 25)

Columns remaining: ['Reference_ID', 'Given_Name', 'Last_Name', 'Major', 'Citizenship', 'Intake', 'Phone_Number', 'Street_1', 'City', 'Region', 'Postal', 'Email_ID', 'Date_of_Birth', 'Most_Recent_Released_Decision', 'Application_Source', 'Is_Global_Grad', 'SEVIS_ID', 'Official_University_email_address', 'Application_Agency_Code', 'Created_At', 'Modified_At', 'SEVIS_ID_missing', 'Reference_ID_capped', 'Created_At_datetime', 'Modified_At_datetime']

EXPLORATORY DATA ANALYSIS (EDA)

Numerical Columns Summary:

	Reference_ID	Recieved_At	Created_At	Modified_At
count	6.871000e+03	6.871000e+03	6.871000e+03	6.871000e+03
mean	1.440063e+06	1.749943e+12	1.749943e+12	1.749956e+12
std	1.190292e+06	7.513078e+08	7.513078e+08	6.593744e+08
min	4.078910e+05	1.740000e+12	1.740000e+12	1.740000e+12
25%	1.409734e+06	1.750000e+12	1.750000e+12	1.750000e+12
50%	1.442069e+06	1.750000e+12	1.750000e+12	1.750000e+12
75%	1.453952e+06	1.750000e+12	1.750000e+12	1.750000e+12
max	1.000050e+08	1.750000e+12	1.750000e+12	1.750000e+12



	SEVIS_ID_missing	Reference_ID_capped
count	6871.00	6871.00
mean	0.62	1428792.28
std	0.48	33069.67
min	0.00	1343408.25
25%	0.00	1409734.50
50%	1.00	1442069.00
75%	1.00	1453952.00
max	1.00	1520278.25

Categorical Columns Summary):

Given_Name:

Fnu: 108 (1.6%)
 Emmanuel: 72 (1.0%)
 Samuel: 46 (0.7%)
 Joseph: 28 (0.4%)
 Michael: 27 (0.4%)

Last_Name:

Mohammed: 115 (1.7%)
 Patel: 83 (1.2%)
 Khan: 40 (0.6%)
 Syed: 34 (0.5%)
 Shaik: 28 (0.4%)

Major:

Information Systems (MS): 1010 (14.7%)

Analytics (MS): 687 (10.0%)

Cybersecurity (MS): 619 (9.0%)

Business Administration (MBA): 574 (8.4%)

Computer Science (MS): 562 (8.2%)

University:

Saint Louis University: 6871 (100.0%)-which is now dropped

Citizenship:

India: 2473 (36.0%)

Nigeria: 1254 (18.3%)

Ghana: 1186 (17.3%)

Zimbabwe: 421 (6.1%)

Kenya: 242 (3.5%)

Intake:

Fall 2025: 4538 (66.0%)

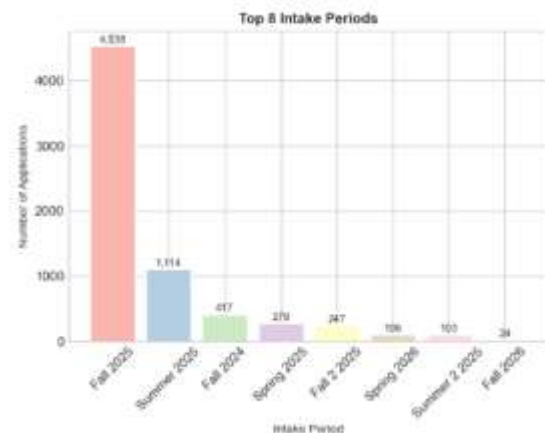
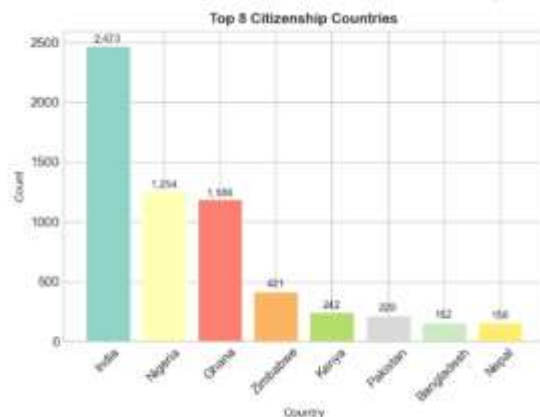
Summer 2025: 1114 (16.2%)

Fall 2024: 417 (6.1%)

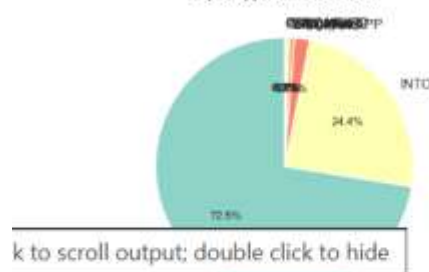
Spring 2025: 279 (4.1%)

Fall 2 2025: 247 (3.6%)

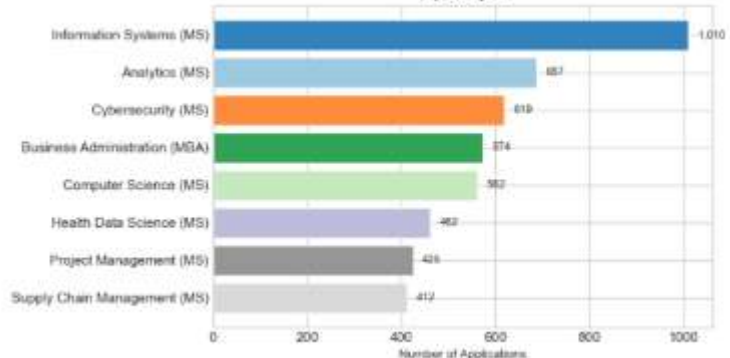
Key Categorical Distributions



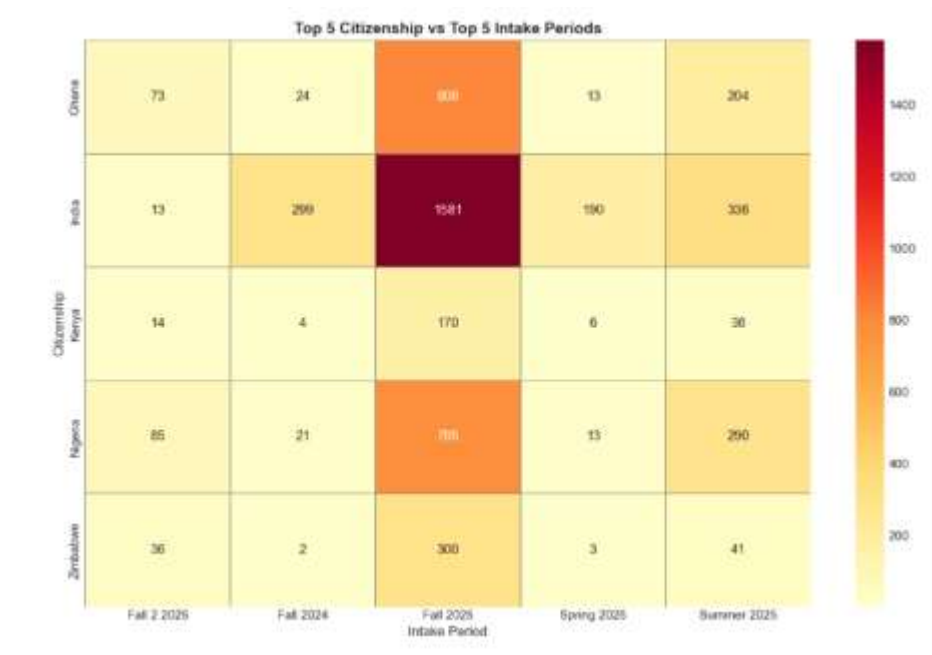
Top 5 Application Sources



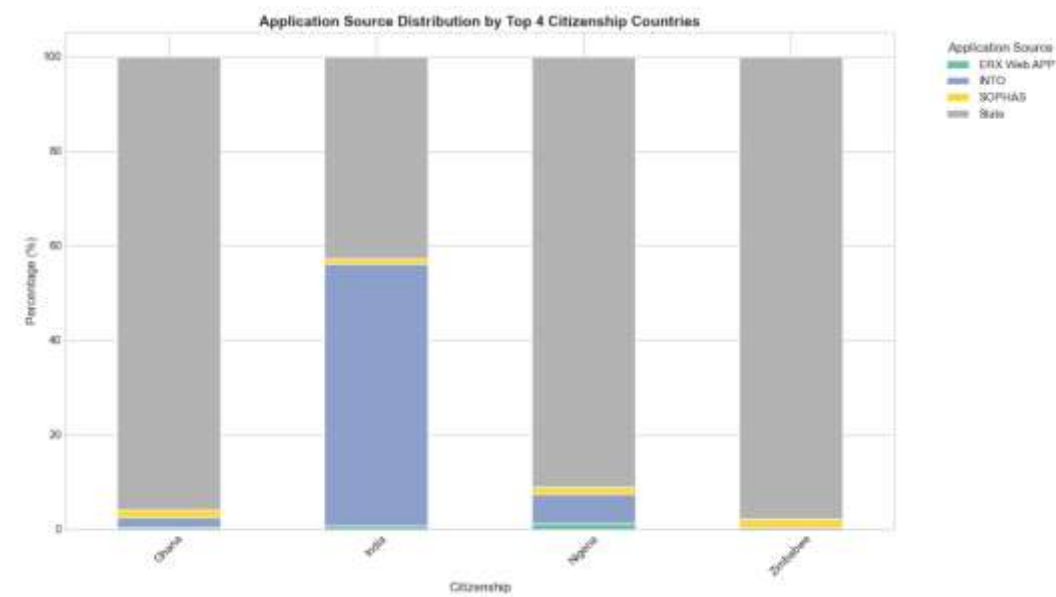
Top 8 Majors



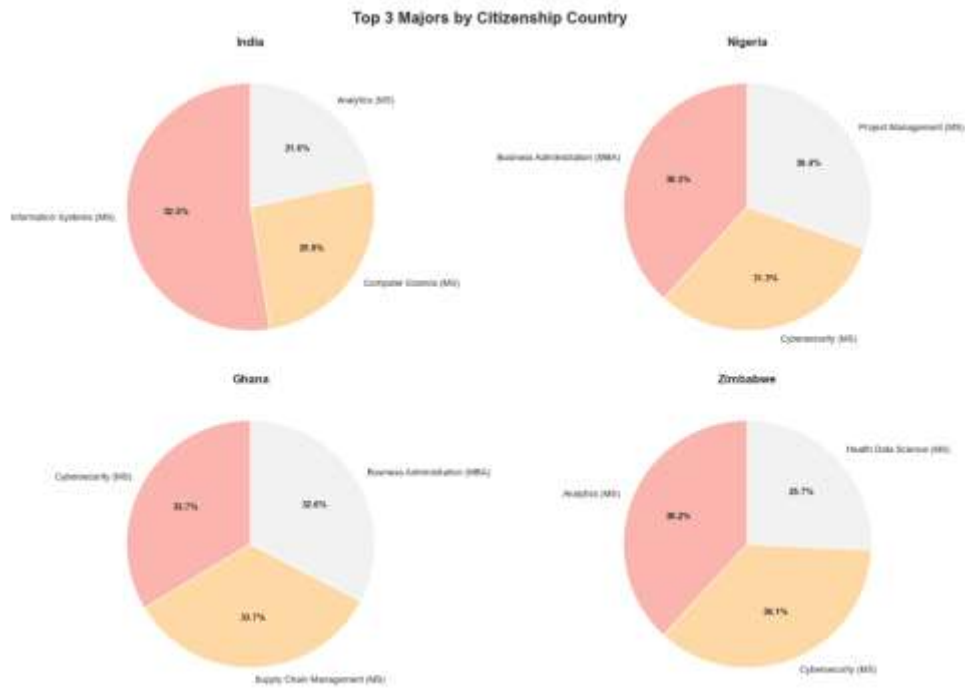
Citizenship vs Intake Heatmap



Application Source Distribution by Top Citizenship



Top Majors by Citizenship - Facet Grid



KEY INSIGHTS & INTERPRETATIONS

1. DATA OVERVIEW:

- Dataset contains 6,871 records with 31 columns
- 6 numerical columns, 22 categorical columns

2. DEMOGRAPHIC INSIGHTS:

- Top applicant country: India (36.0% of total)
- Geographic diversity: 100 different countries
- Most popular major: Information Systems (MS) (14.7% of applications)
- Academic diversity: 88 different majors

3. TEMPORAL PATTERNS:

- Peak intake period: Fall 2025 (66.0% of applications)
- Applications span from 2025-02-19 to 2025-06-15
- Busiest application month: Month 6

4. DATA QUALITY OBSERVATIONS:

- SEVIS ID completeness: 37.8% present
- Reference_ID outliers successfully capped

Report of Connect Dataset

Dataset: 34341 rows × 8 columns

MISSING VALUE ANALYSIS

Total missing values: 19810

Columns with missing values: 1/8

Missing value details:

	Missing_Count	Missing_Percentage
Assigned	19810	57.69
Reference_ID	0	0.00

Deposit_Status	0	0.00
Recieved_At	0	0.00
University	0	0.00
I_20_Status	0	0.00
Created_At	0	0.00
Modified_At	0	0.00

DUPLICATE RECORD ANALYSIS

Full row duplicates: 0 (0.0%)

Duplicate Reference_ID values: 27466

Sample duplicate Reference_IDs:

- 1320577
- 1356800
- 1356915
- 1319991
- 1266321

Top 5 most common assignment patterns:

- 'Catherine Donahue, Prince Tetteh': 1900 times
- 'Catherine Donahue, Palak Joshi': 1170 times
- 'Prince Tetteh': 980 times
- 'Alexander Mikhailov, Prince Tetteh': 620 times
- 'Aaron Greenberg, Prince Tetteh': 480 times

OUTLIER DETECTION

Numeric columns found: ['Reference_ID', 'Created_At']

Outliers detected in 1 columns:

- Reference_ID: 1470 outliers (4.3%)
Range: 407891.00 to 100004958.00, IQR: 1409978.00-1453973.00

EXPLORATORY DATA ANALYSIS (EDA) REPORT

Analyzing cleaned dataset: 34341 rows × 7 columns

Descriptive Statistics:

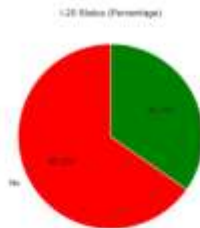
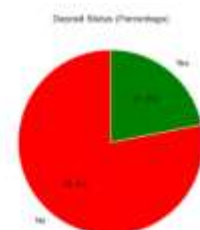
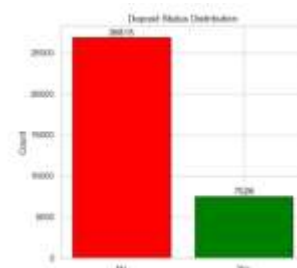
Numeric columns:

	Reference_ID	Created_At
count	3.434100e+04	3.434100e+04
mean	1.440460e+06	1.749994e+12
std	1.190442e+06	2.472161e+08
min	4.078910e+05	1.740000e+12
25%	1.409978e+06	1.750000e+12
50%	1.442138e+06	1.750000e+12
75%	1.453973e+06	1.750000e+12
max	1.000050e+08	1.750000e+12

Categorical columns (value counts):

Deposit_Status:

- No: 26815 (78.1%)
- Yes: 7526 (21.9%)



Assigned:

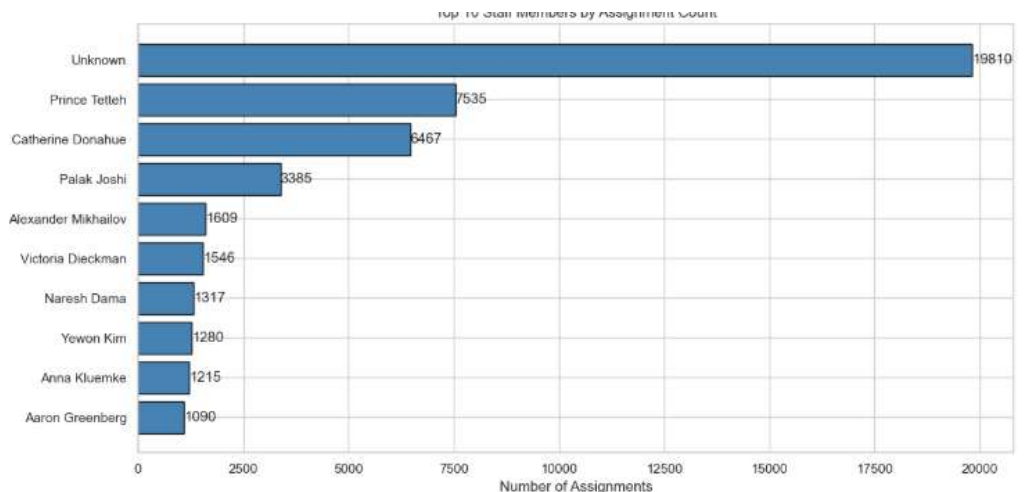
- Unknown: 19810 (57.7%)
- Catherine Donahue, Prince Tetteh: 1900 (5.5%)
- Catherine Donahue, Palak Joshi: 1170 (3.4%)
- Prince Tetteh: 980 (2.9%)
- Alexander Mikhailov, Prince Tetteh: 620 (1.8%)

Recieved_At:

- 5/27/2025 11:58: 6864 (20.0%)
- 5/27/2025 12:00: 6864 (20.0%)
- 5/27/2025 14:05: 6864 (20.0%)
- 5/27/2025 15:02: 6864 (20.0%)
- 5/27/2025 15:47: 6864 (20.0%)

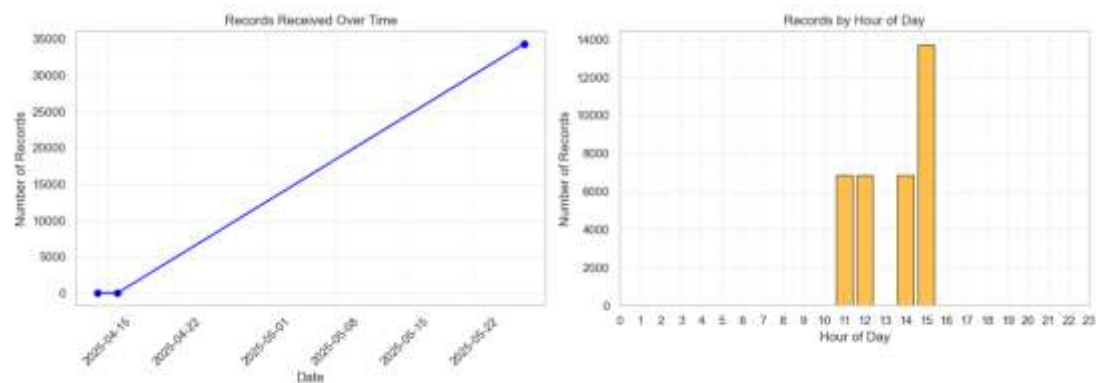
Staff Assignment Analysis:

- Total unique staff involved: 33
- Top 5 staff members:
 - Unknown: 19810 assignments (57.7%)
 - Prince Tetteh: 7535 assignments (21.9%)
 - Catherine Donahue: 6467 assignments (18.8%)
 - Palak Joshi: 3385 assignments (9.9%)
 - Alexander Mikhailov: 1609 assignments (4.7%)

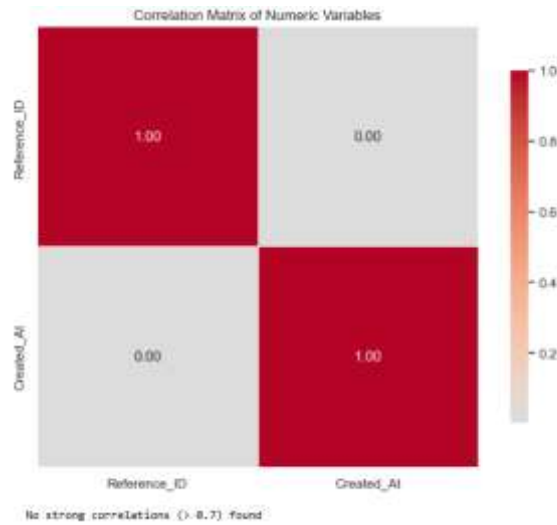


Timeline Analysis:

- Date range: 2025-04-14 to 2025-05-27
- Total days with data: 3
- Busiest day: 2025-05-27 with 34320 records



Correlation Analysis:



KEY INSIGHTS AND INTERPRETATIONS MAJOR FINDINGS:

1. DATA QUALITY:

- Initial data had 19810 missing values
- 0 duplicate rows identified
- After cleaning: 9 columns remain

2. BUSINESS INSIGHTS:

- 21.9% of records have Deposit_Status = 'Yes'
- 34.7% of records have I_20_Status = 'Yes'
- Multiple staff assignments common (team approach)

3. OPERATIONAL PATTERNS:

- Most records received around hour 15:00

4. DATA ISSUES IDENTIFIED:

- 'Recieved_At' column name has typo (should be 'Received_At')
- Scientific notation in Created_At/Modified_At columns
- 'NULL' string used instead of actual NaN values
- Mixed date formats in Recieved_At column